

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Write Characteristics of Digital Signals.	Analyzing	
B)	Explain the working of Digital Gate with their types.	Understanding	
C)	State and prove any two theorem of Boolean algebra..	Applying	
Q.2	Solve Any Two of the following.		12
A)	Explain the working of Multiplexer and De-Multiplexer.	Understanding	
B)	Design a half-adder and full-adder circuits using k-map	Applying	
C)	Minimize the four-variable logic function using k-map. $f(A,B,C,D) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 14)$	Applying	
Q. 3	Solve Any Two of the following.		12
A)	Design 3-bit synchronous up counter using JK flip flops	Applying	
B)	Drew and explain serial in serial out shift register in detail.	Applying	
C)	Write and explain any two applications of flip-flop.	Understanding	
Q.4	Solve Any Two of the following.		12
A)	Comparison of 8-bit, 16-bit, and 32-bit microprocessors.	Understanding	
B)	Draw the pin diagram of 8086 and explain in brief.	Understanding	
C)	Write short note on Memory.	Understanding	
Q. 5	Solve Any Two of the following.		12
A)	Explain different type of Addressing modes of 8086.	Analyzing	
B)	Write short note on assembler and compiler.	Analyzing	
C)	Explain classification of instruction set.	Understanding	

*** End ***